

INFO 2201

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ANALYSING HOUSING VIOLATIONS IN NEW YORK CITY



What is the current state of housing in New York City?



Center for New York City Affairs

New York's Housing Is Falling Apart. Here's How to Stop That.

<https://www.centrernyc.org/urban-matters-2/new-yorks-housing-is-falling-apart-heres-how-to-stop-that>

- 24% of New York City buildings had rodent infestations in 2021
- 17% had cracks in ceilings or floors
- 16% needed more heat in the winter
 - 10% saw heat shut off in the winter, which can lead to dangerous reliance on space heaters or open ovens
- ~16% of buildings with elevators had elevator breakdowns.



Exploring Trends in Housing Equity

The Data

Housing Maintenance Code Violations

NYC OpenData

https://data.cityofnewyork.us/Housing-Development/Housing-Maintenance-Code-Violations/wvxf-dwi5/about_data

Open database of all Housing Violations that get reported to the New York City Government

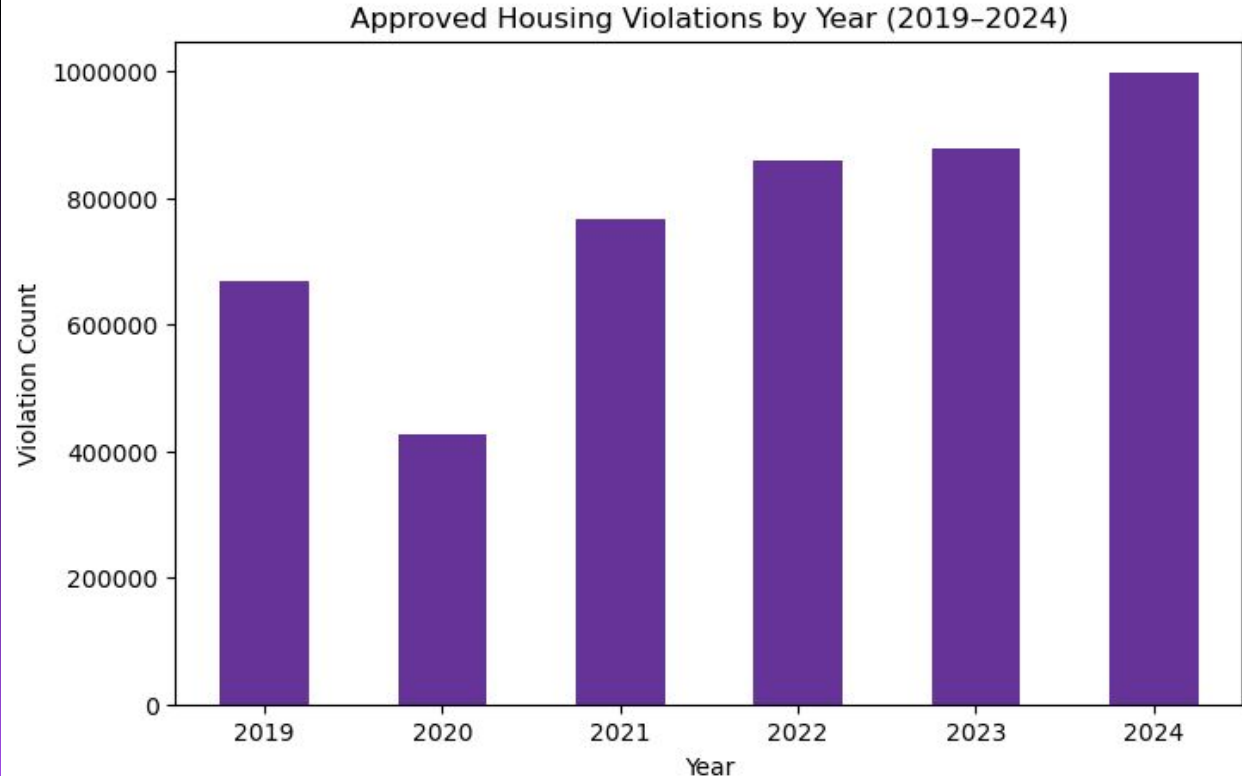
- Specific locations and violations
- Violation details
- Documentation

Cleaning

- Queried Data from NYC OpenData's Website for necessary information
 - Filtered out any information about building addresses
- Because of size, it was important to pull only the columns and years that we were focused on
 - 2019, 2021 (Immediately pre-post COVID)
 - 2024 (Current day)

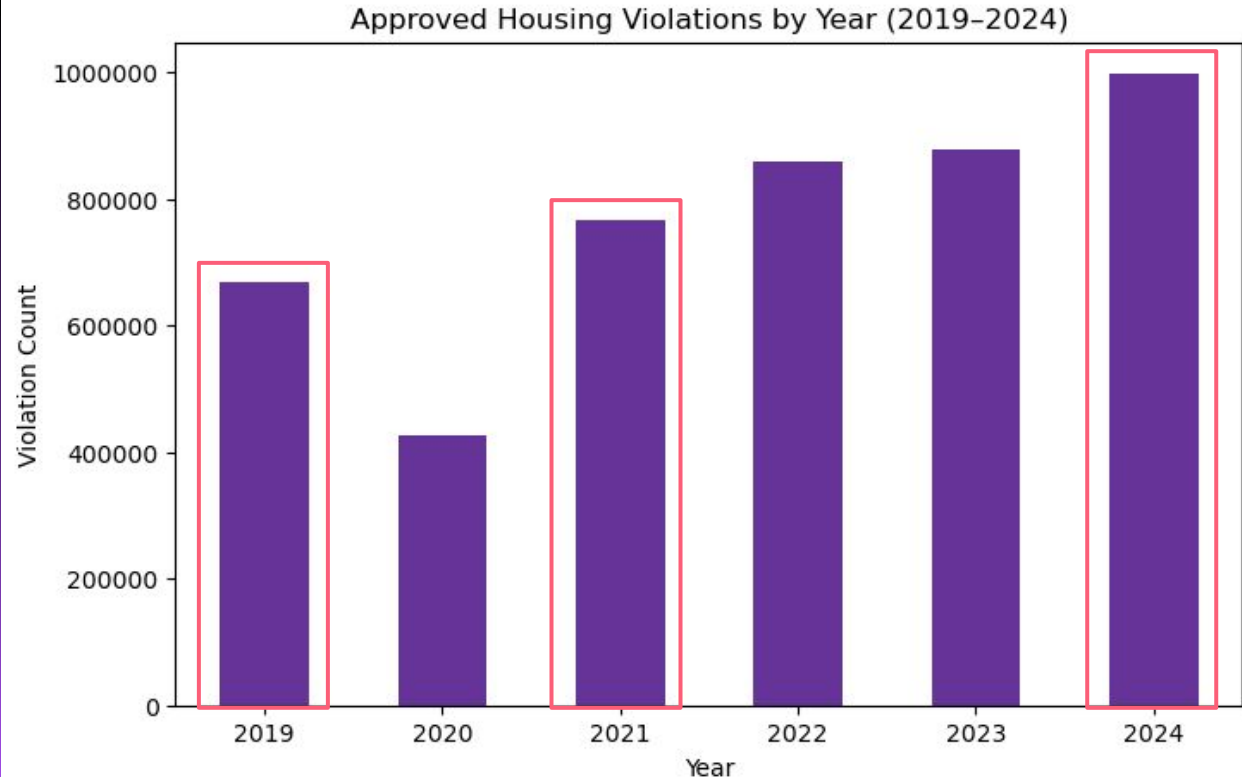
Identifying the Story within the Data

- What Is Unique?
- Are There Any Outliers?
- What Story Does My Data Tell Me?
- 2020 drop was expected likely due to lower enforcement from COVID
- Steady trend upward from COVID
- What Further Research Do I Need To Do To Find The Story?
- Why is there an upward trend in 2024?



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Violation Class Distribution by Year

- Class 'A'

○

- Class 'B'

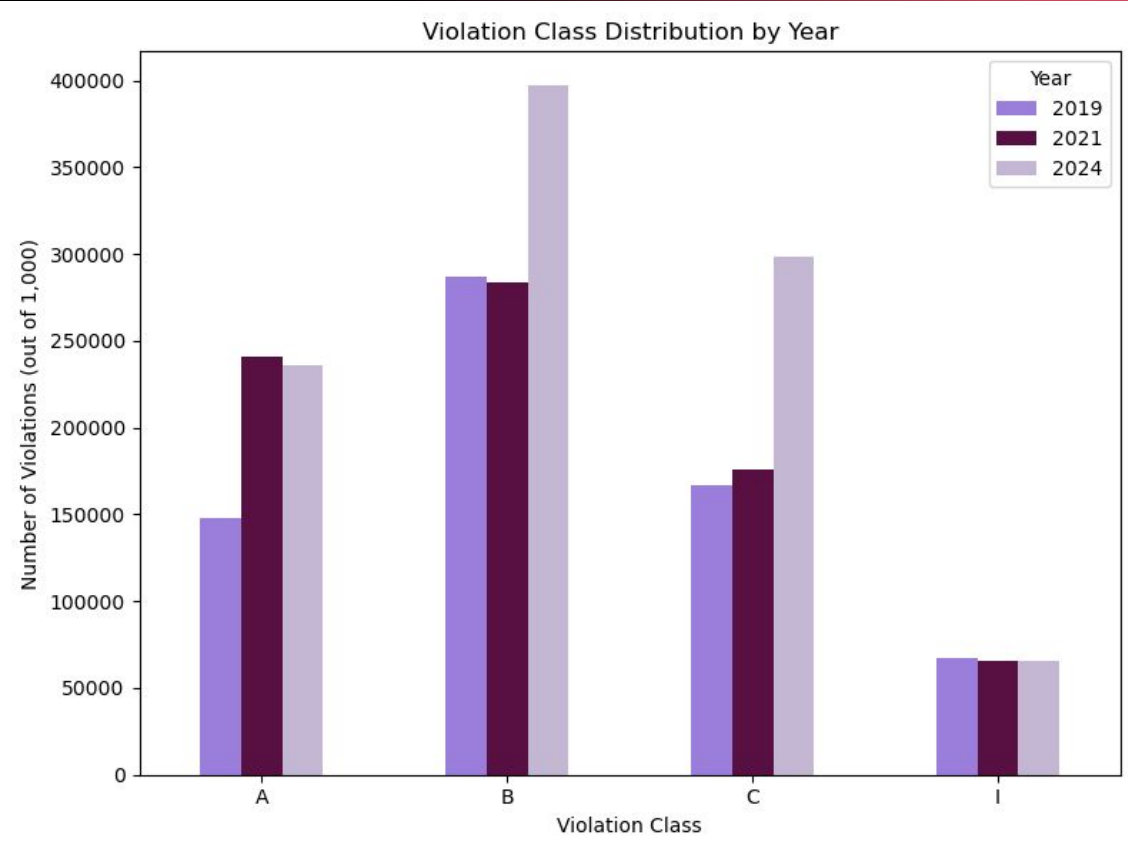
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- Class 'C'

○

- Class 'I'

○





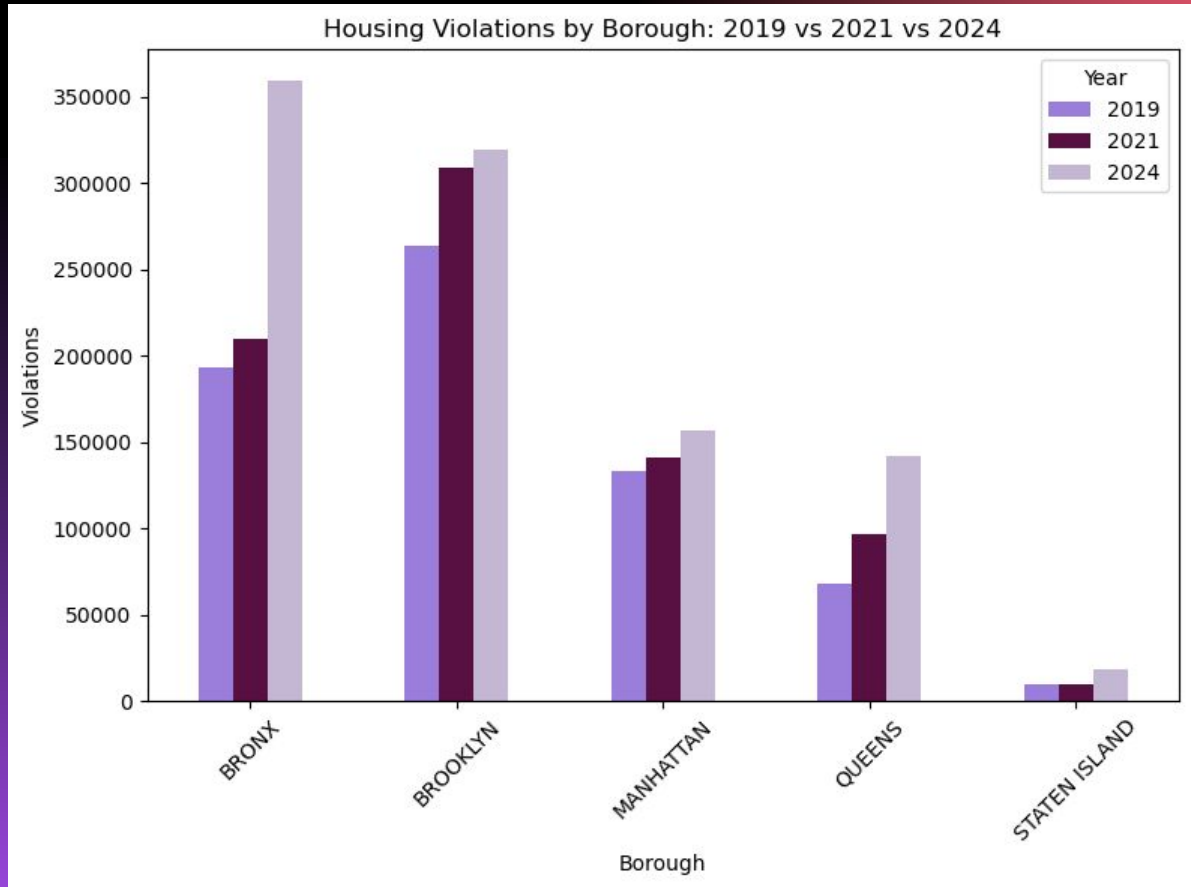
The Boroughs

- Each borough is a county of New York State, though they function primarily as distinct areas within the larger city
- Because each has a unique area of governance, we decided to divide the data up accordingly

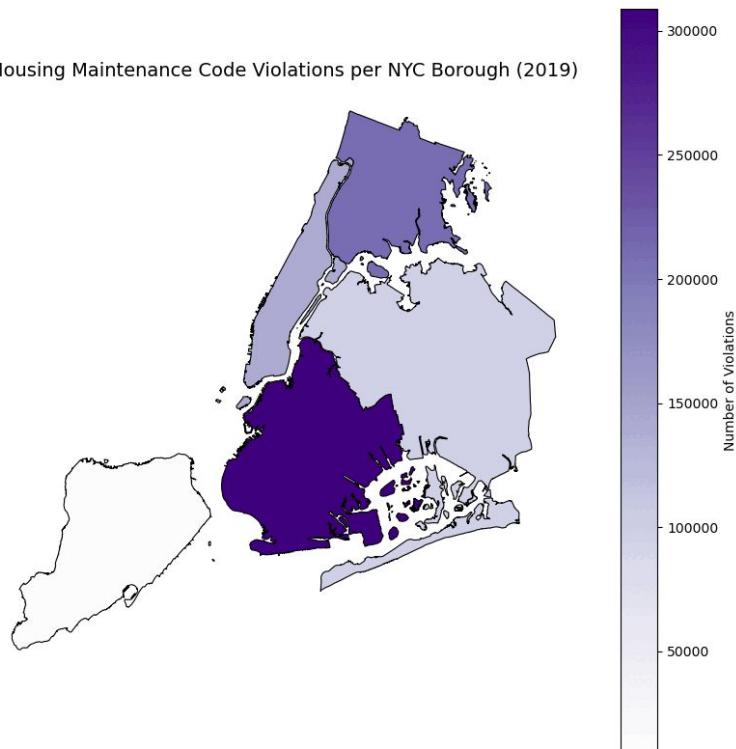
Where are the Violations happening?

Bronx experiences sharp increase in violations in 2024

Brooklyn trends on the higher end



Housing Maintenance Code Violations per NYC Borough (2019)



Spatial Analysis

- Stayed relatively consistent between 2019 and 2021
 - Brooklyn leading Housing Code Violations
 - The Bronx in second
- The Bronx skyrockets in 2024

	BoroName	geometry	Violations
0	Staten Island	MULTIPOLYGON (((970217.022 145643.332, 970227....	10167
1	Queens	MULTIPOLYGON (((1029606.077 156073.814, 102957...	67761
2	Brooklyn	MULTIPOLYGON (((1021176.479 151374.797, 102100...	263624
3	Manhattan	MULTIPOLYGON (((981219.056 188655.316, 980940....	133529
4	Bronx	MULTIPOLYGON (((1012821.806 229228.265, 101278...	193456

2019

2021

	BoroName	geometry	Violations
0	Staten Island	MULTIPOLYGON (((970217.022 145643.332, 970227....	9575
1	Queens	MULTIPOLYGON (((1029606.077 156073.814, 102957...	97103
2	Brooklyn	MULTIPOLYGON (((1021176.479 151374.797, 102100...	308831
3	Manhattan	MULTIPOLYGON (((981219.056 188655.316, 980940....	141208
4	Bronx	MULTIPOLYGON (((1012821.806 229228.265, 101278...	210255

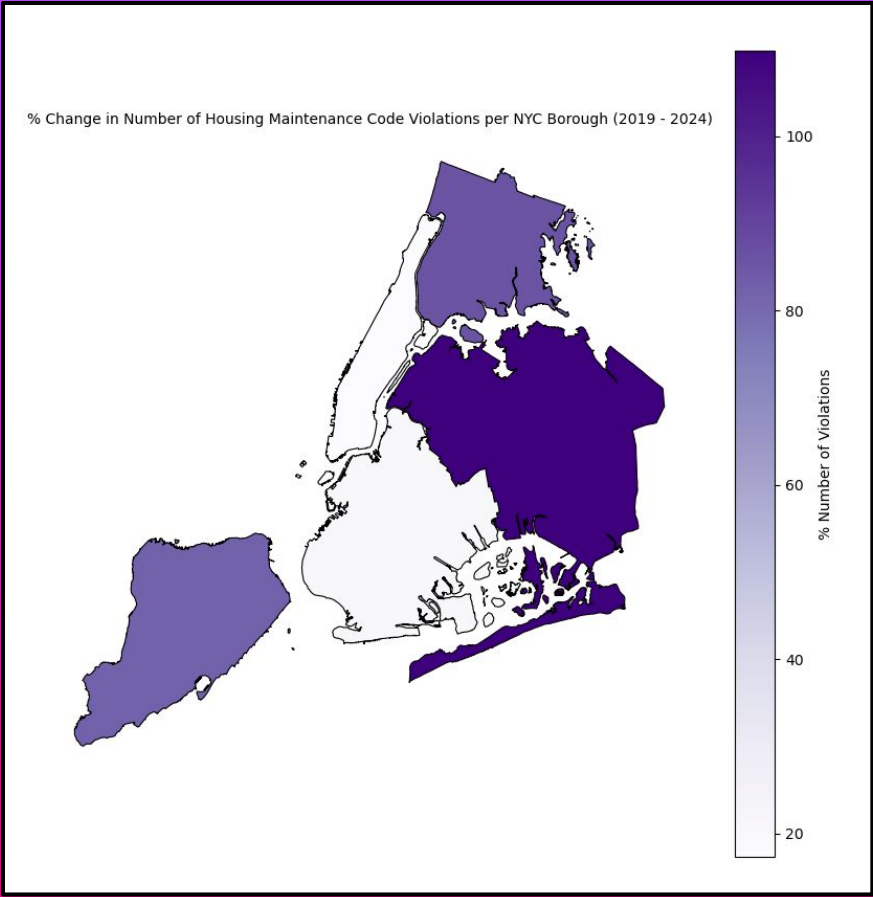
	BoroName	geometry	Violations
0	Staten Island	MULTIPOLYGON (((970217.022 145643.332, 970227....	18514
1	Queens	MULTIPOLYGON (((1029606.077 156073.814, 102957...	142190
2	Brooklyn	MULTIPOLYGON (((1021176.479 151374.797, 102100...	319742
3	Manhattan	MULTIPOLYGON (((981219.056 188655.316, 980940....	156669
4	Bronx	MULTIPOLYGON (((1012821.806 229228.265, 101278...	359466

2024

% Change in Violations per Borough (2019 - 2024)

- The Queens experienced the most percent change in Housing Code Violations
 - Followed by the Bronx and Staten Island
- Housing Violations are increasing in typically lower violation count areas

	BoroName	geometry	pct_change
0	Staten Island	MULTIPOLYGON (((970217.022 145643.332, 970227....	82.098948
1	Queens	MULTIPOLYGON (((1029606.077 156073.814, 102957...	109.840469
2	Brooklyn	MULTIPOLYGON (((1021176.479 151374.797, 102100...	21.287136
3	Manhattan	MULTIPOLYGON (((981219.056 188655.316, 980940....	17.329569
4	Bronx	MULTIPOLYGON (((1012821.806 229228.265, 101278...	85.812795



CITYLIMITS

Housing Violations in NYC Jumped 24% This Year. We Mapped Them By Neighborhood.

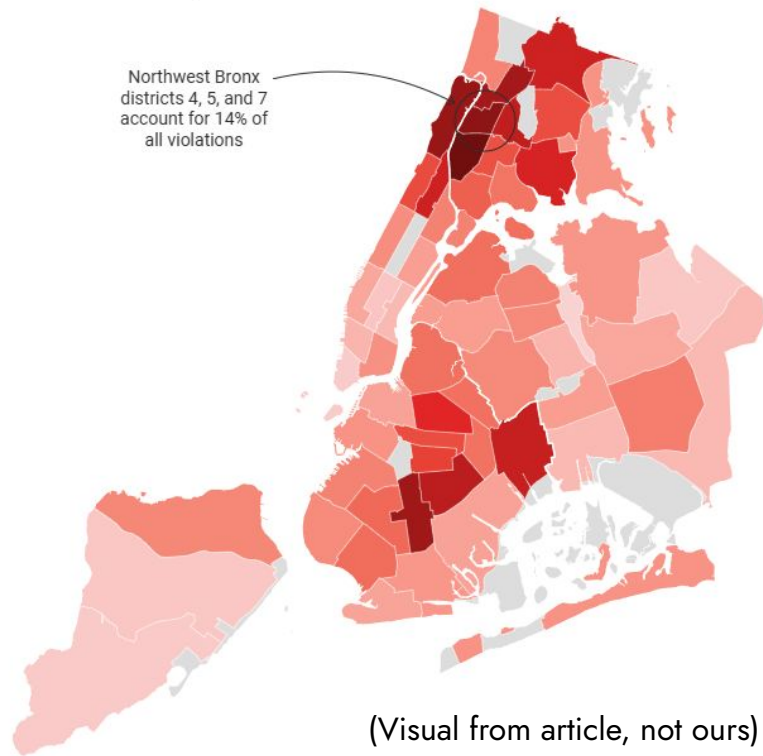
BY PATRICK SPAUSTER OCTOBER 16, 2024 4 MIN

<https://citylimits.org/housing-violations-in-nyc-jumped-24-this-year-we-mapped-them-by-neighborhood/>

- Bronx and Southeast Brooklyn are leaders in Housing Violations
- This is a big change that took place within one year with the Bronx overtaking Southeast Brooklyn
- We have found that our findings could correlate with information found in this article.

The Bronx and Southeast Brooklyn Have More Housing Maintenance Violations

Housing maintenance violations between July 2023 and June 2024 by community district



CITYLIMITS

The Bronx Twin Park Fire

- Many new violations stemmed from the Bronx Twin Park fire of 2022 which was started by a faulty space heater.
- Leading to a 48% increase in heat violations, as well as over 73,000 violations related to malfunctioning interior doors and public spaces.
- One building fell into such disrepair that 396 violations were reported in a two year span.



What is the current state of housing in New York City?



- Rent stabilized buildings do not provide enough income to cover operating expenses
 - Gap could be subsidized by city government
 - Under Mayor Eric Adams, city has opted to defund subsidized housing
- Leaders in housing violations like the Bronx and Brooklyn hold the highest populations of Black and Hispanic peoples
 - Clear racial inequity in the city
- As enforcement increases, more and more violations become reported
 - Large trends like in the Bronx highlight a need for increased enforcement

THANK YOU

Any questions?

Bibliography

Evelly, W. by J. E. J. (2024, October 18). *Housing violations in NYC jumped 24% This year. we mapped them by neighborhood*. City Limits.

<https://citylimits.org/housing-violations-in-nyc-jumped-24-this-year-we-mapped-them-by-neighborhood/>

Moncrease, Seth. "New York's Housing Is Falling Apart. Here's How to Stop That." *Urban Matters*, Center for New York City Affairs, The New School, January 11, 2023.

<https://www.centrernyc.org/urban-matters-2/new-yorks-housing-is-falling-apart-heres-how-to-stop-that>

City of New York Department of Housing Preservation and Development. "Housing Maintenance Code Violations." *NYC Open Data*. Metadata updated July 19, 2025. Accessed July 21, 2025.

https://data.cityofnewyork.us/Housing-Development/Housing-Maintenance-Code-Violations/wvxf-dwi5/about_data